Lecture 7. Optimization

Golden section, Newton, line search, Fisher scoring, GLM and IRLS, Gauss–Newton, gradient and coordinate descent

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We consider $\min_{x\in\mathbb{R}^d} f(x)$; maximization is minimization of $-f$.

7.1 Golden section search

For $d = 1$, let f be continuous and **strictly unimodal** on $[a, b]$: $f(\lambda x+(1-\lambda)y) < \max\{f(x), f(y)\}$ for $x \neq y$, $\lambda \in (0, 1)$. Keep three points $a < x_1 < b$ with $f(x_1) < \min\{f(a), f(b)\}$, so the minimum lies in (a, b). Place a new point x_0 in the longer of (a, x_1) and (x_1, b); say $a < x_0 < x_1$.

- If $f(x_0) > f(x_1)$, the minimum is in (x_0, b): set $(a, x_1, b) \leftarrow (x_0, x_1, b)$.
- Otherwise it is in (a, x_1): set $(a, x_1, b) \leftarrow (a, x_0, x_1)$.

To make both outcomes equally good and keep the configuration self-similar, require $b-x_0 = x_1-a$ and that the interior point divides the new interval in the same proportion. With $\alpha = (x_0-a)/(b-a)$ and $y = (x_1 - x_0)/(b - a)$, this gives $\alpha + y + \alpha = 1$ and $1 : \alpha = (y + \alpha) : y$, whose solution is

$$\alpha = \frac{3-\sqrt{5}}{2}, \qquad 1-\alpha = \frac{\sqrt{5}-1}{2} \approx 0.618,$$

the golden section. Each step shrinks the interval by the factor 0.618 with one function evaluation: slow (linear) but guaranteed to find the global minimum of a unimodal function.

7.2 Newton's method

At a minimizer, $\nabla f(x^\star) = 0$. Applying Newton–Raphson to ∇f is equivalent to minimizing the local quadratic model

$$f(x) \approx f(x^{(t)}) + \nabla f(x^{(t)})^\top (x - x^{(t)}) + \tfrac{1}{2}(x - x^{(t)})^\top \nabla^2 f(x^{(t)})(x - x^{(t)}),$$

giving the **naive Newton** iteration

$$x^{(t+1)} = x^{(t)} - [\nabla^2 f(x^{(t)})]^{-1} \nabla f(x^{(t)}).$$

It converges quadratically near a minimizer with positive definite Hessian. But it is not a descent method: it is equally happy to go uphill and can converge to a maximum or a saddle, or diverge, depending on the start (try $f = \sin$).

7.3 Practical Newton: positive definite approximations and line search

Motivation. Pure Newton steps for optimization can move uphill when the Hessian is not positive definite, or overshoot far from the optimum. Modifying the Hessian and choosing the step length by line search keep the fast local convergence while guaranteeing descent.

Replace the Hessian by a positive definite $H^{(t)}$ and choose a step size s:

$$x^{(t+1)} = x^{(t)} + s\,\Delta x^{(t)}, \qquad \Delta x^{(t)} = -[H^{(t)}]^{-1} \nabla f(x^{(t)}).$$

Proposition 7.1 (Descent direction)

If $H^{(t)}$ is positive definite and $\nabla f(x^{(t)}) \neq 0$, then $f(x^{(t)} + s\Delta x^{(t)}) < f(x^{(t)})$ for all sufficiently small $s > 0$.

Proof. By Taylor's theorem, $f(x^{(t)}+s\Delta x^{(t)})-f(x^{(t)}) = -s\,\nabla f^\top [H^{(t)}]^{-1}\nabla f + o(s)$, and the quadratic form is positive. ■

($\{\nabla f^\top H^{-1}\nabla f\}^{1/2}$ is the **Newton decrement**.) For strictly convex f the true Hessian already works (**damped Newton**), but a line search is still needed for global convergence. Taking $H^{(t)} = I$ gives gradient descent.

Line search. Given the direction, choose s: exactly (minimize $f(x+s\Delta x)$, e.g. by golden section), or approximately by step halving or **backtracking** (Armijo): with $\alpha \in (0, \frac{1}{2})$ and $\beta \in (0,1)$, start from $s = 1$ and set $s \leftarrow \beta s$ while

$$f(x + s\Delta x) > f(x) + \alpha s \nabla f(x)^\top \Delta x.$$

The condition accepts s once f lies below a line whose slope is a fraction α of the initial slope, which always happens for small s by Proposition 7.1. The direction is computed once per iteration; the line search costs only function evaluations. A common implementation slip: if the trial point $x_s = x + s\Delta x$ is formed first, the right side of the test is $f(x) + \alpha\nabla f(x)^\top (x_s - x)$, which already contains s; writing $\alpha s \nabla f(x)^\top (x_s - x)$ counts s twice and makes the test too lenient. (For IRLS, the trial point for step s can be computed by refitting the weighted least squares with working response $z = \eta + sW^{-1}(y-\mu)$, since the fit is linear in z and returns exactly $\beta + s\Delta\beta$.)

7.4 Fisher scoring

For maximum likelihood, $f = -\ell(\theta)$. **Fisher scoring** replaces $-\nabla^2\ell$ by the expected information

$$I(\theta) = E[-\nabla^2\ell(\theta)] = E[\nabla\ell(\theta)\nabla\ell(\theta)^\top] \succeq 0,$$

(the equality holds when differentiation and expectation can be interchanged), giving

$$\theta^{(t+1)} = \theta^{(t)} + s[I(\theta^{(t)})]^{-1}\nabla\ell(\theta^{(t)}).$$

Example 7.1 (Logistic regression)

$y_i \sim \mathrm{Ber}(p_i)$ with $\operatorname{logit} p_i = \eta_i = x_i^\top\beta$. Then

$$\ell(\beta) = \sum_i \{y_i x_i^\top \beta - \log(1 + e^{x_i^\top\beta})\}, \qquad \nabla\ell = X^\top(y-p),$$

$$-\nabla^2\ell = X^\top W X, \qquad W = \mathrm{diag}\{p_i(1-p_i)\}.$$

The Hessian does not involve y, so $I(\beta) = -\nabla^2\ell$: Newton and Fisher scoring coincide. The update rearranges into

$$\beta^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} z^{(t)}, \qquad z^{(t)} = X\beta^{(t)} + s(W^{(t)})^{-1}(y - p^{(t)}),$$

a weighted least squares fit to the **working responses** $z^{(t)}$: **iteratively reweighted least squares** (IRLS). Any weighted least squares solver therefore fits logistic regression.

Example 7.2 (Poisson regression)

$y_i \sim \mathrm{Poi}(\lambda_i)$ with $\log\lambda_i = x_i^\top\beta$: $\ell(\beta) = \sum_i\{y_i x_i^\top\beta - e^{x_i^\top\beta} - \log y_i!\}$, $\nabla\ell = X^\top(y-\lambda)$, $-\nabla^2\ell = X^\top W X$ with $W = \mathrm{diag}(\lambda_i)$. Again the Hessian is free of y, and Newton = Fisher scoring = IRLS with $z = X\beta + W^{-1}(y-\lambda)$.

7.5 Generalized linear models

These coincidences are no accident. An **exponential dispersion family** has density

$$p(y \mid \eta, \phi) = \exp\Big\{\frac{y\eta - b(\eta)}{a(\phi)} + c(y, \phi)\Big\},$$

with natural parameter η, mean $\mu = b'(\eta)$ and variance $b''(\eta)a(\phi)$. A **GLM** links the mean to a linear predictor, $g(\mu) = x^\top\beta$; the **canonical link** is $g = (b')^{-1}$, which makes $\eta = x^\top\beta$.

family	canonical link	variance function
normal	$\eta = \mu$	1
Poisson	$\eta = \log\mu$	μ
binomial	$\eta = \log\frac{\mu}{1-\mu}$	$\mu(1-\mu)$
gamma	$\eta = \mu^{-1}$	μ^2
inverse Gaussian	$\eta = \mu^{-2}$	μ^3

Proposition 7.2

With $a(\phi) \equiv 1$, $\mu_i = g^{-1}(x_i^\top\beta)$ and $\sigma_i^2 = b''(\eta_i)$:

$$\nabla\ell(\beta) = \sum_i \frac{(y_i - \mu_i)}{\sigma_i^2 g'(\mu_i)} x_i, \qquad I(\beta) = \sum_i \frac{1}{\sigma_i^2 g'(\mu_i)^2} x_i x_i^\top = X^\top W X.$$

Fisher scoring is IRLS with weights $w_i = 1/\{\sigma_i^2 g'(\mu_i)^2\}$. For the canonical link, $-\nabla^2\ell = I(\beta)$, so Newton = Fisher scoring = IRLS and $-\ell$ is convex.

Proof. $\ell_i = y_i\eta_i - b(\eta_i)$, so $\partial\ell_i/\partial\eta_i = y_i - \mu_i$. By the chain rule, $d\mu = b''(\eta)d\eta = \sigma^2 d\eta$ and $d\mu = x^\top d\beta / g'(\mu)$, so $\partial\eta_i/\partial\beta = x_i/\{\sigma_i^2 g'(\mu_i)\}$, which gives the score. Differentiating again, one term is $-x_i x_i^\top/\{\sigma_i^2 g'(\mu_i)^2\}$ and the other is proportional to $(y_i - \mu_i)$, which has mean 0; this gives $I(\beta)$. For the canonical link $\eta_i = x_i^\top\beta$, so $\partial\eta_i/\partial\beta = x_i$ and $\nabla^2\ell = -\sum_i b''(\eta_i) x_i x_i^\top$, which involves no y_i and equals $-I(\beta)$ (here $\sigma_i^2 g'(\mu_i) = 1$). ■

For non-canonical links (e.g. probit) Fisher scoring and Newton differ and the likelihood need not be concave (probit happens to be log-concave). R's `glm` uses Fisher scoring.

Example 7.3 (Probit regression)

Let $y_i \sim \mathrm{Ber}(p_i)$ with $p_i = \Phi(\eta_i)$, $\eta_i = \beta_0 + \beta_1 x_i$, and write $\Phi_i = \Phi(\eta_i)$, $\phi_i = \phi(\eta_i)$, $\mathbf{x}_i = (1, x_i)^\top$. Then

$$\ell(\beta) = \sum_i \{y_i \log\Phi_i + (1 - y_i)\log(1 - \Phi_i)\}, \qquad \nabla\ell = \sum_i \frac{(y_i - \Phi_i)\phi_i}{\Phi_i(1 - \Phi_i)}\mathbf{x}_i.$$

Differentiating ϕ_i/Φ_i and $-\phi_i/(1 - \Phi_i)$ in η with $\phi'(z) = -z\phi(z)$ gives

$$\nabla^2\ell = \sum_i f_i\,\mathbf{x}_i\mathbf{x}_i^\top, \qquad f_i = -y_i\frac{\phi_i(\eta_i\Phi_i + \phi_i)}{\Phi_i^2} - (1 - y_i)\frac{\phi_i\{\phi_i - \eta_i(1 - \Phi_i)\}}{(1 - \Phi_i)^2},$$

which depends on y_i: the link is not canonical. Taking expectations ($Ey_i = \Phi_i$) gives the Fisher information $X^\top W X$ with $w_i = \phi_i^2/\{\Phi_i(1 - \Phi_i)\}$, in agreement with Proposition 7.2 ($\sigma_i^2 = \Phi_i(1 - \Phi_i)$, $g'(\mu_i) = 1/\phi_i$). The IRLS step is

$$\beta^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} z^{(t)}, \qquad z_i^{(t)} = \eta_i^{(t)} + \frac{y_i - \Phi_i^{(t)}}{\phi_i^{(t)}}.$$

7.6 Nonlinear least squares: Gauss–Newton

Gauss located the dwarf planet Ceres in 1801 by least squares. In general, minimize

$$f(\beta) = \tfrac{1}{2}\sum_{i=1}^{n}\{y_i - \mu_i(\beta)\}^2,$$

e.g. a logistic growth curve $\mu(x;\beta) = \beta_3/(1+e^{-\beta_1-\beta_2 x})$. With $J(\beta)$ the $n \times p$ Jacobian whose rows are $\nabla\mu_i(\beta)^\top$,

$$\nabla f = -J^\top(y-\mu), \qquad \nabla^2 f = J^\top J - \sum_i (y_i - \mu_i)\nabla^2\mu_i.$$

Gauss–Newton drops the second term, using the positive semidefinite $J^\top J$:

$$\beta^{(t+1)} = \beta^{(t)} + s(J^\top J)^{-1}J^\top(y - \mu(\beta^{(t)})),$$

i.e. each step is a linear least squares fit of the residuals on J. It is justified when residuals are small or μ is nearly linear, and it is exactly Fisher scoring for the model $y_i \sim N(\mu_i(\beta), \sigma^2)$, since $E[-\nabla^2\ell] \propto J^\top J$. (Gauss introduced the normal distribution partly to justify least squares.)

7.7 Gradient descent

$$x^{(t+1)} = x^{(t)} - \gamma_t \nabla f(x^{(t)}).$$

Minimizing the linear model $\nabla f^\top \Delta x$ over $\|\Delta x\|_2 \le 1$ gives $\Delta x = -\nabla f/\|\nabla f\|$: the steepest descent direction. Because a linear model is unbounded below, a step size must always be chosen. Each iteration is cheap and needs no Hessian, which makes the method attractive at large scale; but it zigzags and converges slowly, and it does not apply to nonsmooth f.

Theorem 7.3

Let f be convex and differentiable with L-Lipschitz gradient, $\|\nabla f(x) - \nabla f(y)\|_2 \le L\|x-y\|_2$, and let f attain its minimum $p^\star$ at $x^\star$. With constant step $\gamma \in (0, 1/L]$,

$$f(x^{(t)}) - p^\star \le \frac{\|x^{(0)} - x^\star\|_2^2}{2\gamma t}.$$

Proof. The Lipschitz gradient gives the quadratic upper bound $f(y) \le f(x) + \nabla f(x)^\top(y-x) + \frac{L}{2}\|y-x\|^2$. With $y = x^+ = x - \gamma\nabla f(x)$ and $\gamma L \le 1$, $f(x^+) \le f(x) - \frac{\gamma}{2}\|\nabla f(x)\|^2$ (descent). By convexity, $f(x) \le p^\star + \nabla f(x)^\top(x - x^\star)$, so

$$f(x^+) - p^\star \le \nabla f(x)^\top(x-x^\star) - \frac{\gamma}{2}\|\nabla f(x)\|^2 = \frac{1}{2\gamma}(\|x-x^\star\|^2 - \|x^+ - x^\star\|^2).$$

Summing over t iterations telescopes: $\sum_{k=1}^{t}\{f(x^{(k)}) - p^\star\} \le \frac{1}{2\gamma}\|x^{(0)} - x^\star\|^2$. Since $f(x^{(k)})$ is nonincreasing, the left side is at least $t\{f(x^{(t)}) - p^\star\}$. ∎

This $O(1/t)$ rate is *sublinear*; with strong convexity one gets linear convergence, but never Newton's quadratic rate.

Example 7.4

- Least squares $f(\beta) = \frac{1}{2}\|X\beta - y\|^2$: $\nabla f = X^\top X\beta - X^\top y$ is Lipschitz with $L = \sigma_{\max}(X)^2$, so $\gamma \le 1/\sigma_{\max}(X)^2$ works.

- Logistic regression: $\nabla f = -X^\top(y-p)$ is Lipschitz with $L = \frac{1}{4}\|X^\top X\|_2$, since $p(1-p) \le \frac{1}{4}$. The minimum may not exist (e.g. separable data); a ridge penalty $\frac{\rho}{2}\|\beta\|^2$ fixes this.
- Poisson regression: the gradient $-X^\top(y - e^{X\beta})$ is not globally Lipschitz, so a constant step size can fail; line search is needed. Indeed, an L-Lipschitz gradient forces $v^\top \nabla^2 f(\beta) v \le L\|v\|^2$ for all β, v. But for any v with $x_j^\top v > 0$ for some j,

$$v^\top \nabla^2 f(tv) v = \sum_i e^{t x_i^\top v} (x_i^\top v)^2 \ge e^{t x_j^\top v} (x_j^\top v)^2 \to \infty \quad (t \to \infty).$$

Example 7.5 (Multinomial logistic regression)

With K classes, dummy-coded responses $y_i \in \{0,1\}^K$, and $\log(p_{ik}/p_{iK}) = x_i^\top \beta_k$ ($k < K$, $\beta_K = 0$), we have $p_{ik} = e^{x_i^\top \beta_k}/(1 + \sum_{l<K} e^{x_i^\top \beta_l})$. With $B = [\beta_1, \dots, \beta_{K-1}] \in \mathbb{R}^{p\times(K-1)}$, Y_{-K} the first $K-1$ columns of Y, and P the $n \times (K-1)$ matrix of the p_{ik},

$$\ell(B) = \sum_i \sum_k y_{ik} \log p_{ik} = \operatorname{tr}(Y_{-K}^\top X B) - \sum_i \log\Big(1 + \sum_{l<K} e^{x_i^\top \beta_l}\Big),$$

$$\nabla_{\beta_k} \ell = \sum_i x_i (y_{ik} - p_{ik}), \qquad \text{so} \qquad B^{(t+1)} = B^{(t)} + \gamma X^\top (Y_{-K} - P^{(t)})$$

is gradient ascent. For $K = 2$ it reduces to logistic regression. Each step costs $O(npK)$ and needs no $p(K-1)$-dimensional Hessian; the gradient is Lipschitz (the Hessian is bounded by $\frac{1}{2}\|X\|_2^2$ in operator norm), so a constant step works.

Example 7.6 (Quantiles by optimization)

The α-quantile of X (cdf F, positive density f) solves $F(x) = \alpha$. Newton's method is $x^{(t+1)} = x^{(t)} - \{F(x^{(t)}) - \alpha\}/f(x^{(t)})$. It is Newton's method for minimizing a function ψ with $\psi' = F - \alpha$ and $\psi'' = f > 0$. Since $F(x) = E[\mathbf{1}\{X \le x\}]$, integrating in x and interchanging with the expectation gives

$$\psi(x) = E\big[(x - X)\mathbf{1}\{X \le x\}\big] - \alpha x = E\big[\rho_\alpha(X - x)\big] + \text{const}, \qquad \rho_\alpha(u) = u\{\alpha - \mathbf{1}(u < 0)\},$$

the expected **check loss** of quantile regression; it is convex. Gradient descent reads $x^{(t+1)} = x^{(t)} - \gamma_t\{F(x^{(t)}) - \alpha\}$ and needs no density. If F is unavailable but X is easy to sample, replace the gradient by its unbiased one-sample estimate,

$$x^{(t+1)} = x^{(t)} - \gamma_t(\mathbf{1}\{X_t \le x^{(t)}\} - \alpha), \qquad X_t \overset{\text{iid}}{\sim} F,$$

with $\sum_t \gamma_t = \infty$, $\sum_t \gamma_t^2 < \infty$ (e.g. $\gamma_t = c/t$): this *stochastic gradient* (Robbins–Monro) algorithm converges to the quantile almost surely. The simplest Monte Carlo alternative is the sample α-quantile of N draws, which has standard error $\sqrt{\alpha(1-\alpha)/N}/f(x_\alpha)$.

7.8 Coordinate descent

Minimize over one coordinate (or block) at a time, cycling:

$$x_j^{(t+1)} = \arg\min_{x_j} f(x_1^{(t+1)}, \dots, x_{j-1}^{(t+1)}, x_j, x_{j+1}^{(t)}, \dots, x_d^{(t)}).$$

Each one-dimensional problem is often easy (sometimes closed form). The objective values decrease monotonically, so they converge. It is the optimization analogue of Gauss–Seidel. But even for

convex f, if f is not differentiable, coordinate descent can get stuck at a non-optimal point where no single coordinate move helps (a "corner" of a nonsmooth convex function).